Aman Singh

SOFTWARE ENGINEER INTERN

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SUMMARY

Results-oriented Computer Science undergraduate specializing in **Full-Stack Software Engineering, AI Backend Systems, and Unity Game Development**. Proven track record of building **5 production-grade applications** in **C++, Python, and C#**, cutting API retrieval latency by **35%**, and solving **320+ algorithmic problems** on LeetCode and CodeChef. Seeking a Software Engineer Intern role to architect scalable, high-performance systems.

WORK EXPERIENCE

ABES Engineering College — AI & Backend Engineering Lab — **Ghaziabad, India**

***Software Engineer Intern (RAG Question Answering System)* | github.com/not-Lucifer** — *Jan 2026 - Mar 2026*

- Architected a Retrieval-Augmented Generation (RAG) system in **Python and LangChain** to index **50+ academic PDFs** with automated page citations, reducing out-of-scope hallucinations by **40%**.
- Engineered **3 modular vector retrieval pipelines** (Cosine Similarity, MMR, and Hybrid BM25 + ChromaDB) supporting OpenAI and local **LLaMA** models, boosting top-5 recall accuracy to **88%**.
- Deployed a RESTful **FastAPI backend** with a Streamlit UI and **15+ automated pytest unit tests**, benchmarking retrieval quality via **Hit@k and MRR** metrics across **100+ test queries**.

Open-Source Software Development — GameStash Desktop Client — **Ghaziabad, India**

***C++ Software Developer* | github.com/not-Lucifer** — *Oct 2025 - Jan 2026*

- Developed a native Windows desktop app in **C++ and WebView2** aggregating live free-game deals across **4 digital storefronts** (Steam, Epic Games, GOG), cutting manual search time by **75%**.
- Secured local user accounts and claim histories in **SQLite** using **256-bit PBKDF2-HMAC-SHA256** password hashing and **OAuth 2.0 (PKCE)** authentication with **0 security vulnerabilities**.
- Automated CI/CD release packaging via **CMake and Inno Setup** to ship versioned Windows installers on **GitHub Releases** with a **1-click** update check, reducing deployment effort by **60%**.

FitVision Capstone Initiative (Agile Team of 5) — **Ghaziabad, India**

***Frontend & Computer Vision Developer* | github.com/not-Lucifer** — *Aug 2025 - Nov 2025*

- Spearheaded a **5-member agile team** to build a real-time posture tracking web app using **Python, OpenCV, and MediaPipe**, tracking repetitions across **5 compound exercises** at **30 FPS** with LLM voice coaching.
- Revamped the default **Streamlit frontend** using custom CSS layouts to improve user navigation speed by **30%**, and co-authored a **6-page IEEE-format research paper** on pose estimation accuracy.

PROJECTS

Haunted House — 3D Stealth Action Game (Unity 6, C#, URP, Cinemachine) — *Jun 2026 - Oct 2026*

- Expanded Unity's 3D Stealth framework by designing **2 interconnected levels**, multi-stage progression logic, and spatial audio cues, shipping cross-platform builds for **Windows and WebGL** at **60 FPS**.
- Programmed autonomous enemy AI patrol systems using **waypoint navigation, trigger vision cones, and Physics Raycast** line-of-sight checks, reducing false-positive player detections by **25%**.

Kitchen Chaos — Time-Management Game (Unity, C#, Input System, ScriptableObjects) — *Mar 2025 - Jul 2025*

- Constructed a 3D arcade cooking game using a decoupled, **event-driven C# architecture** and **10+ modular ScriptableObject** data containers, reducing inter-class code coupling by **40%**.
- Implemented interactive station state machines, dynamic order queues, and runtime-rebindable controls across **2 input schemes** (Keyboard and Gamepad), cutting input response latency by **20%**.

EDUCATION

ABES Engineering College (Affiliated to AKTU) — **Ghaziabad, Uttar Pradesh**

Bachelor of Technology in Computer Science and Engineering* | *CGPA: 7.88/10.0 — *Aug 2024 - May 2028*

St. Thomas Senior Secondary School — **Uttar Pradesh**

Senior Secondary (Class XII), Science Stream* | *Percentage: 85.8% — *Apr 2021 - Mar 2023*

SKILLS

- **Programming Languages:** C++, C, C#, Python, SQL, HTML5, CSS3
- **Backend & AI Frameworks:** FastAPI, REST APIs, LangChain, ChromaDB, Retrieval-Augmented Generation (RAG), SQLite, Streamlit, OpenCV, MediaPipe
- **Game Development:** Unity 6, C# Scripting, Universal Render Pipeline (URP), Cinemachine, Unity Input System, ScriptableObjects, UI Toolkit
- **Core Computer Science & Developer Tools:** Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks, Git, GitHub, CI/CD, CMake, Pytest, OAuth 2.0, Linux, Visual Studio, VS Code
- **Soft Skills & Engineering Practices:** Agile Scrum Methodology, System Design, Technical Documentation, Code Review, Cross-Functional Collaboration, Analytical Problem Solving

AWARDS & CERTIFICATIONS

- **Competitive Programming (LeetCode & CodeChef):** Solved **320+ algorithmic problems** in C++ (**140+ on LeetCode** with **1461 Contest Rating** & **100 Days Badge 2026**; **182 on CodeChef** across **11 rated contests**).
- **Cisco Networking Academy (Feb 2026):** Certified in Python Essentials 1 & 2, Networking Basics, and Cisco Packet Tracer.
- **NASSCOM FutureSkills Prime & Unity Learn (Oct 2026):** Certified in Exploratory Data Analysis (EDA) and Unity 3D Game Development.